

MATHEMATICS ESSENTIAL

3

EXTERNALLY SET TASK
GENERAL

Total time for the task: 50 minutes
 Total marks: 33 marks
 Weighting: 15% of the school mark

A formula sheet for use in this task is provided on page 11.

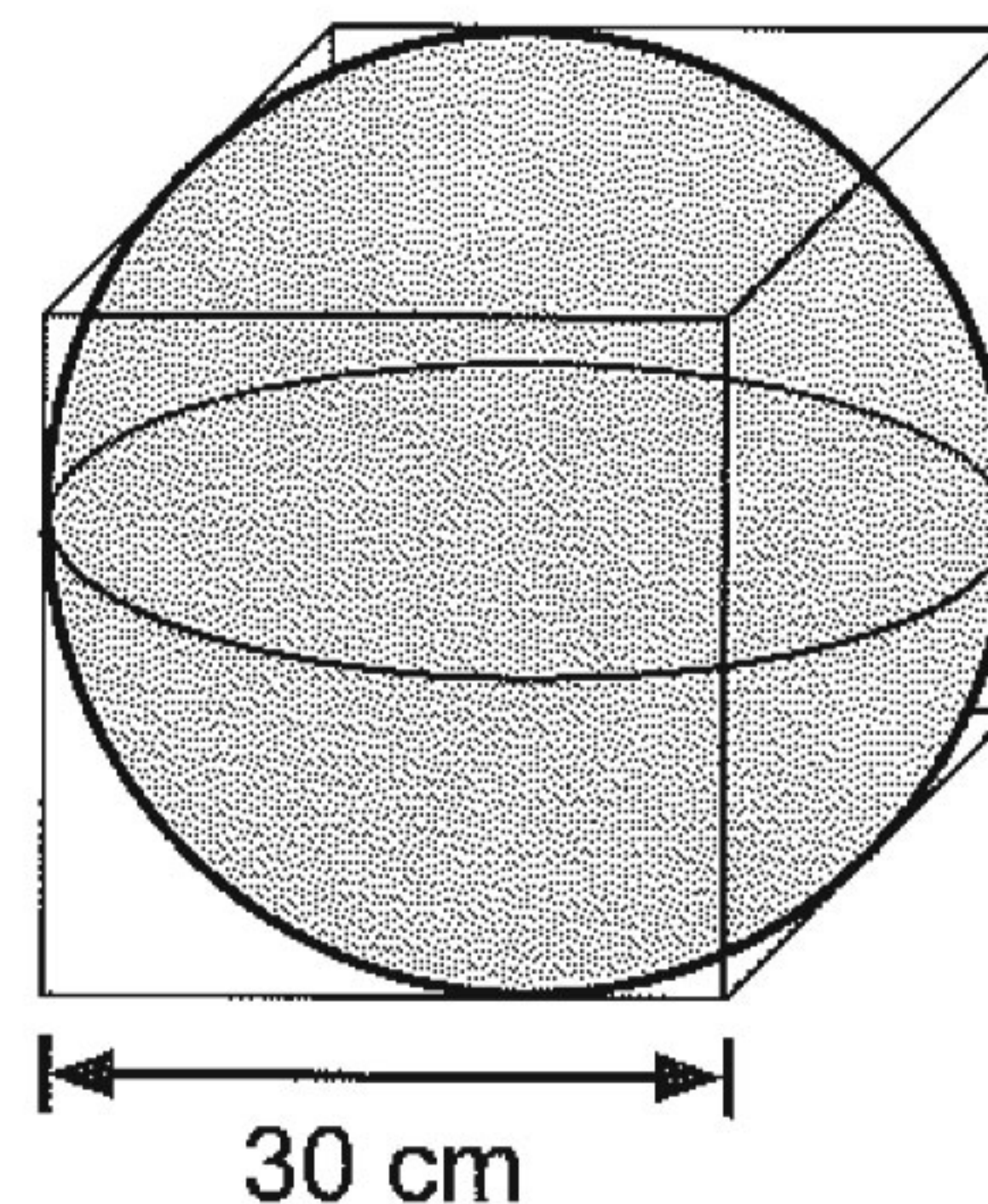
Question 1

(7 marks)

A ball manufacturer makes and delivers balls to various retail shops. The balls are inflated and individually packaged before being boxed for delivery.

The following diagram shows the ball in its individual packaging.

Packaging A

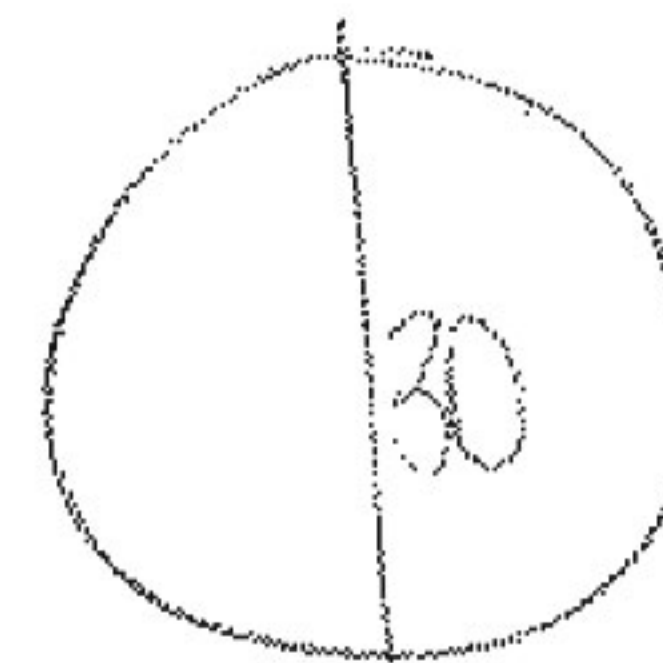


For each ball, determine:

- (a) the volume of the ball to the nearest cm^3 .

(4 marks)

$$\begin{aligned} & \frac{4}{3}\pi r^3 \\ & = \frac{4}{3}\pi (15)^3 \\ & = 14137.17 \\ & = 14137 \text{ cm}^3 \end{aligned}$$



- (b) the amount of air, to the nearest litre (L) that would be required to inflate (fill) the ball.

(2 marks)

$$\begin{aligned} & 14137 \div 1000 = 14.137 \\ & 14.137 \approx 14 \text{ L} \end{aligned}$$

- (c) To avoid damage during delivery, the empty spaces between the ball and its packaging needs to be filled with packaging pellets. Given the volume of the box is $27\,000 \text{ cm}^3$, determine the volume of space that needs to be filled.

(1 mark)

$$\begin{aligned} & 27000 - 14137 \\ & = 12863 \text{ cm}^3 \end{aligned}$$

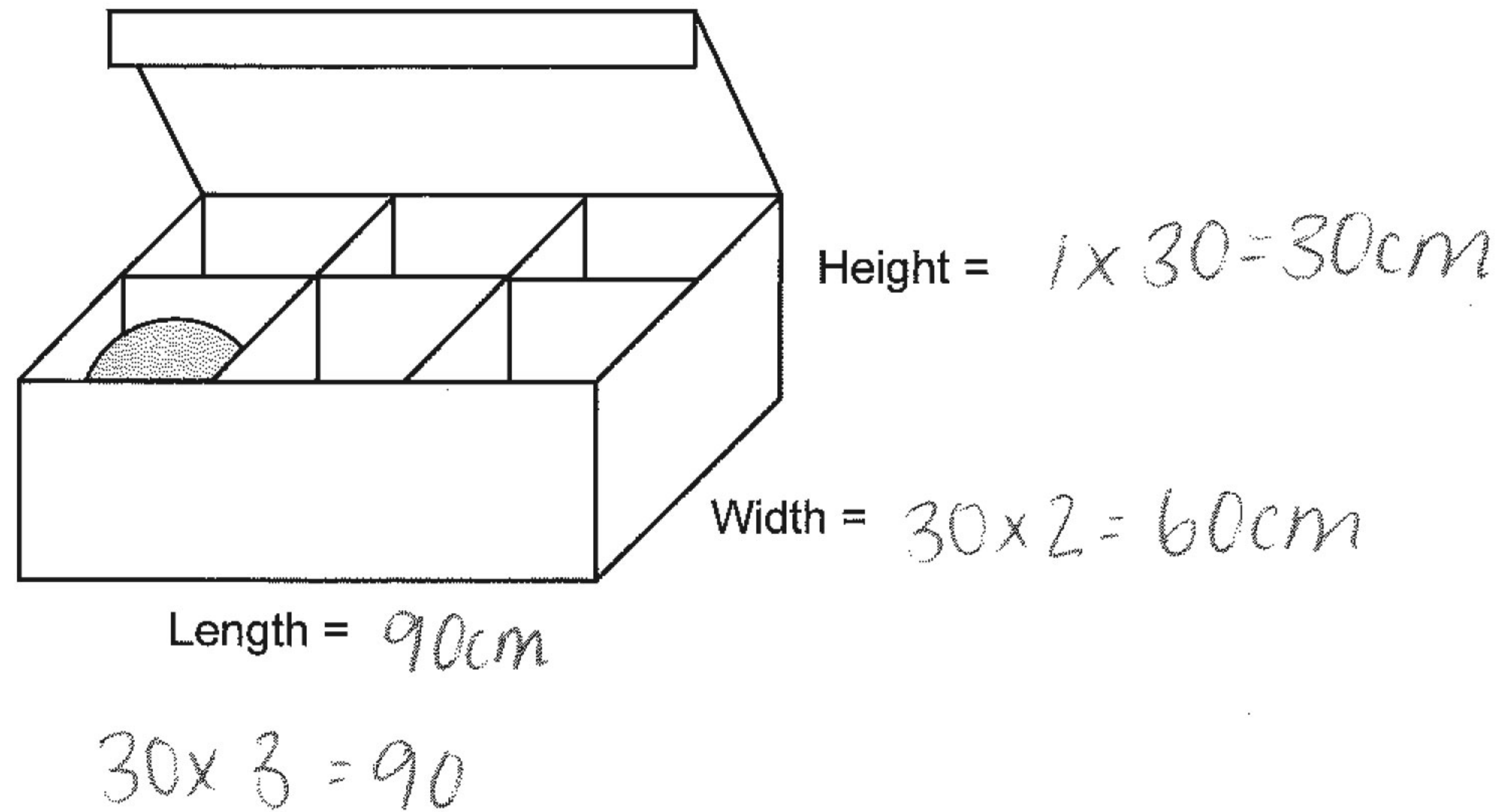
Question 2

(3 marks)

For delivery purposes, the company packs **six** balls (in their packaging) into boxes.

Insert the correct dimensions, in cm, for the length, width and height of each of these boxes.

Box A

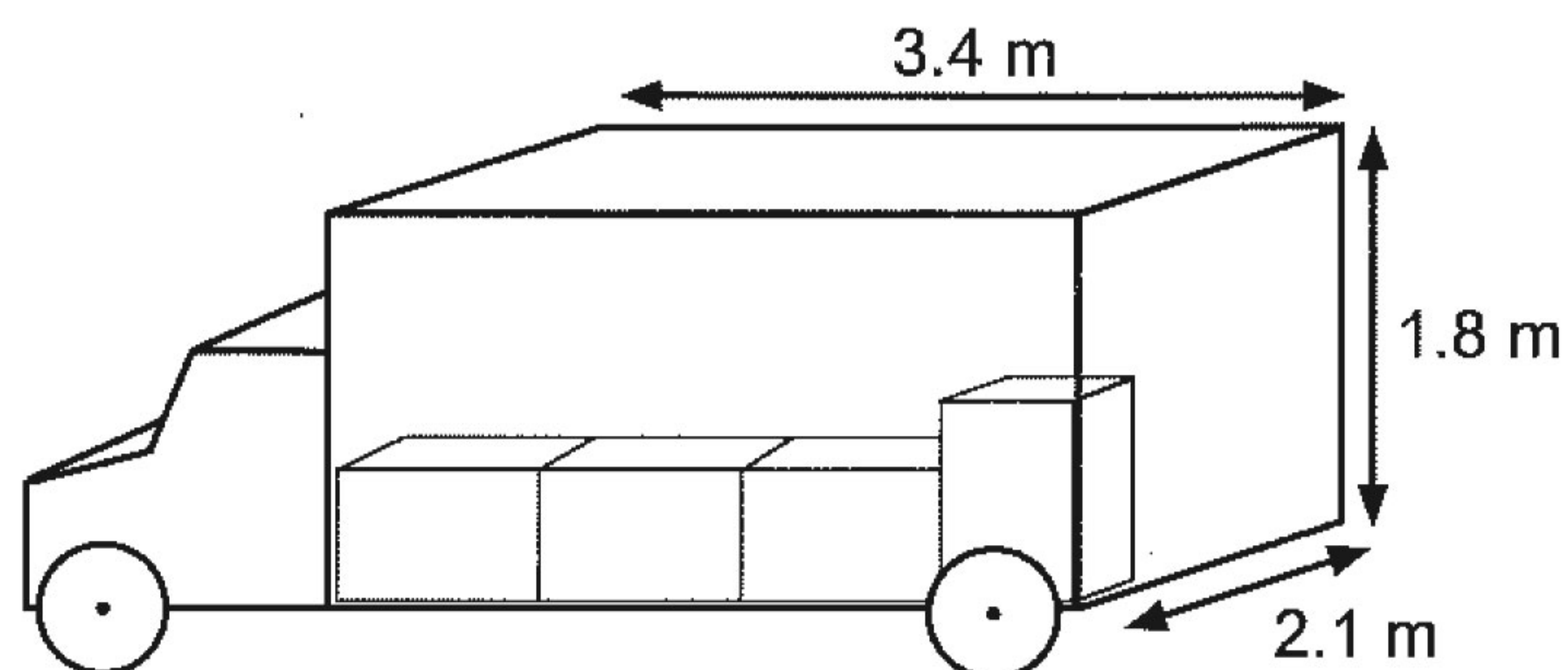


Question 3

(9 marks)

The following diagram shows the dimensions of the delivery truck.

The truck driver begins loading the truck with Box As for delivery, as shown in the diagram below.



12585200
 $162000 = 1$

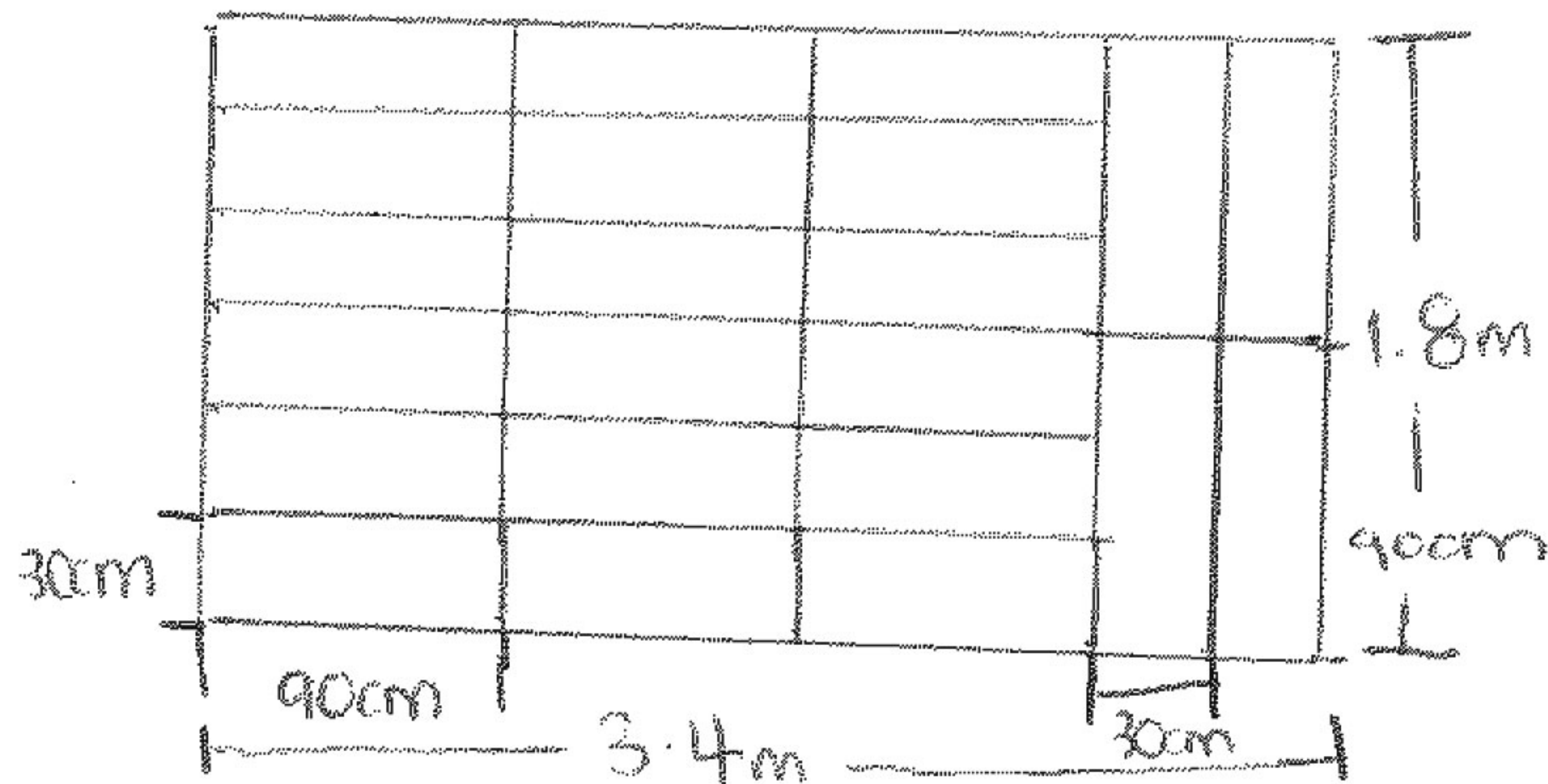
If the driver continues to load the truck until he has placed the maximum number of boxes into the truck:

- (a) sketch the **side**, **top** and **back** views of the truck, showing the dimensions of the boxes inside the truck. (7 marks)

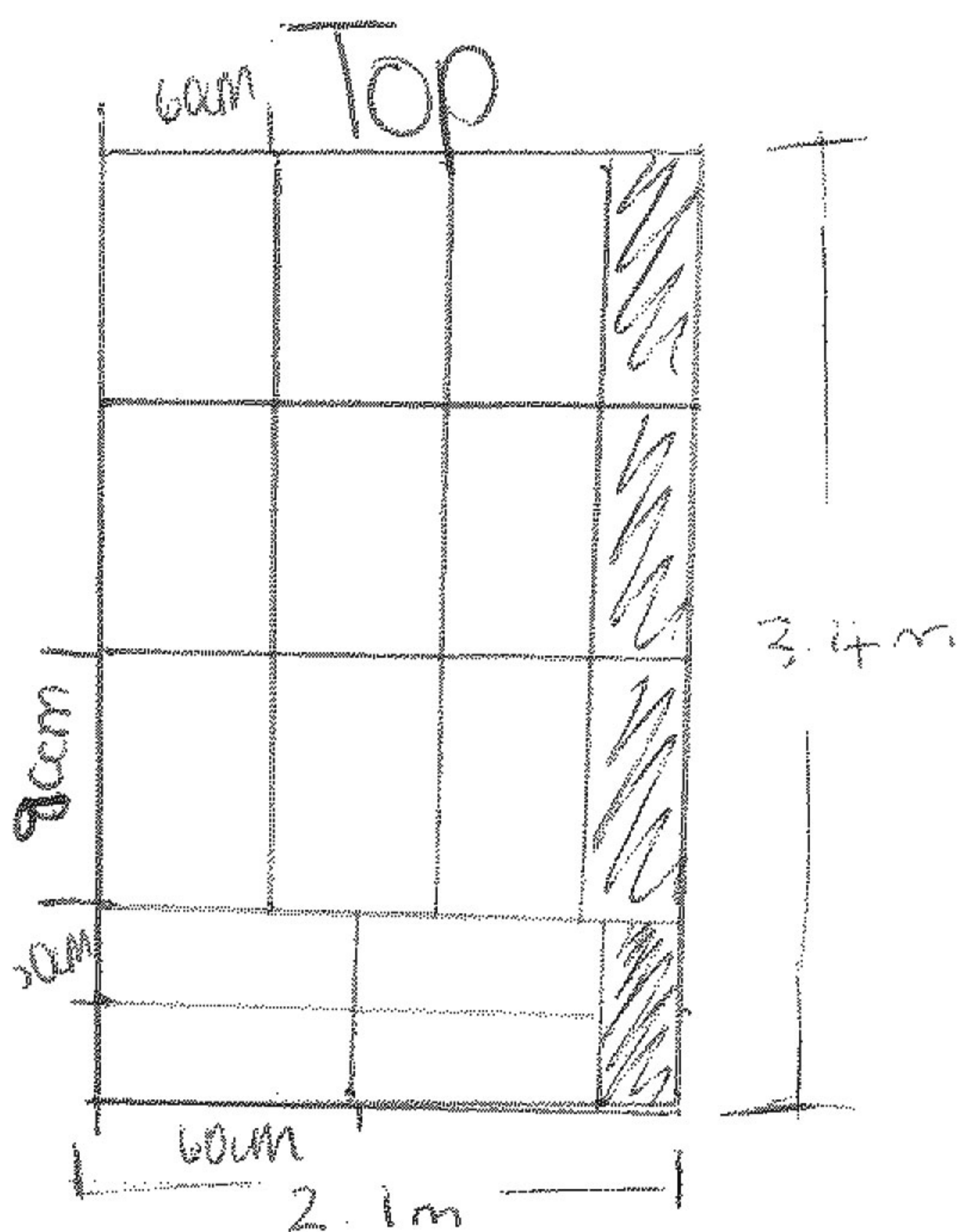
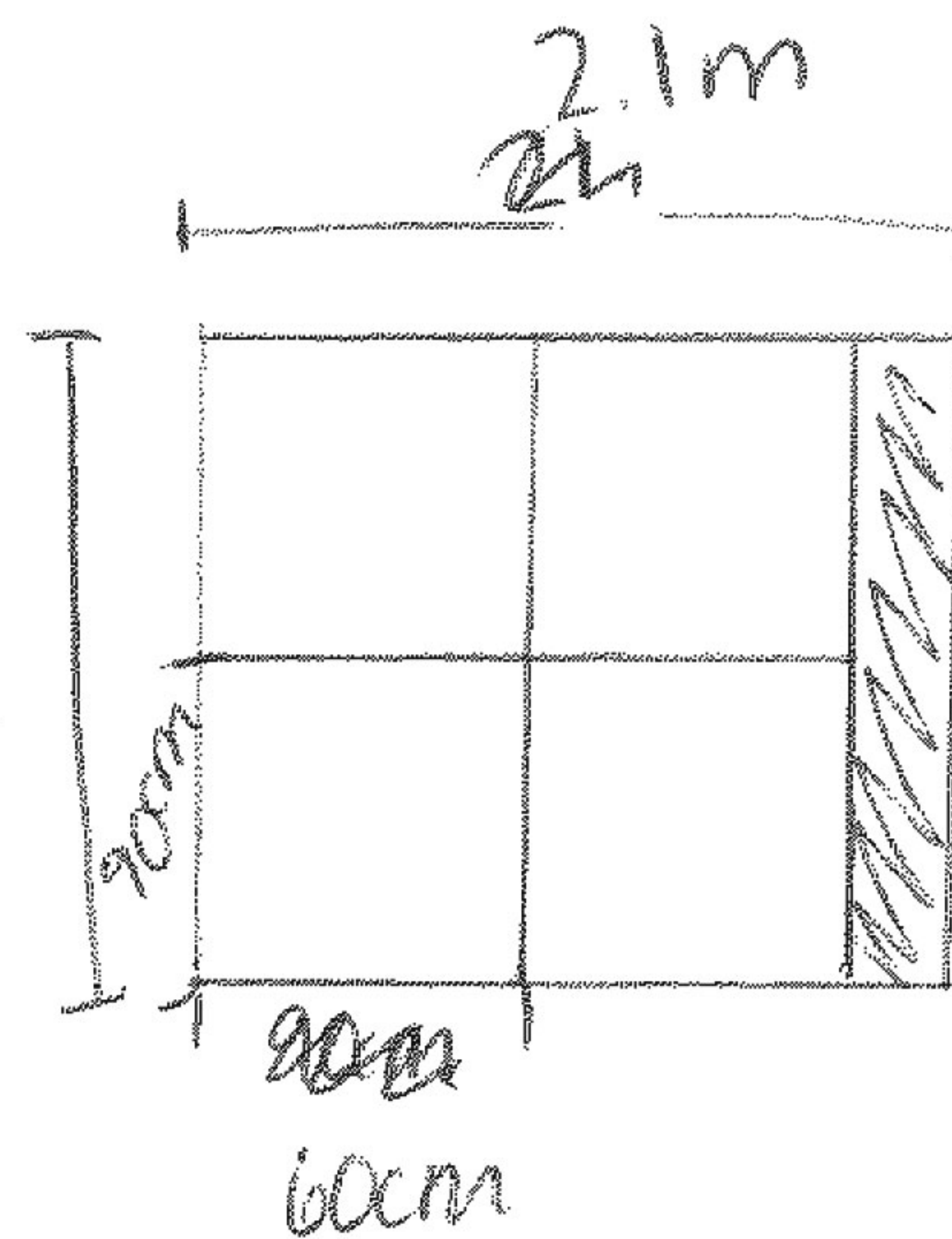
Back Not to Scale

Not to scale

Side



Back



 = empty space

- (b) determine the maximum number of Box As that could fit into the truck. Show your working clearly to justify your answer. (2 marks)

$$L \times W \times H$$
$$3.4 \times 2.1 \times 1.8 = 12.852 = \text{Truck}$$

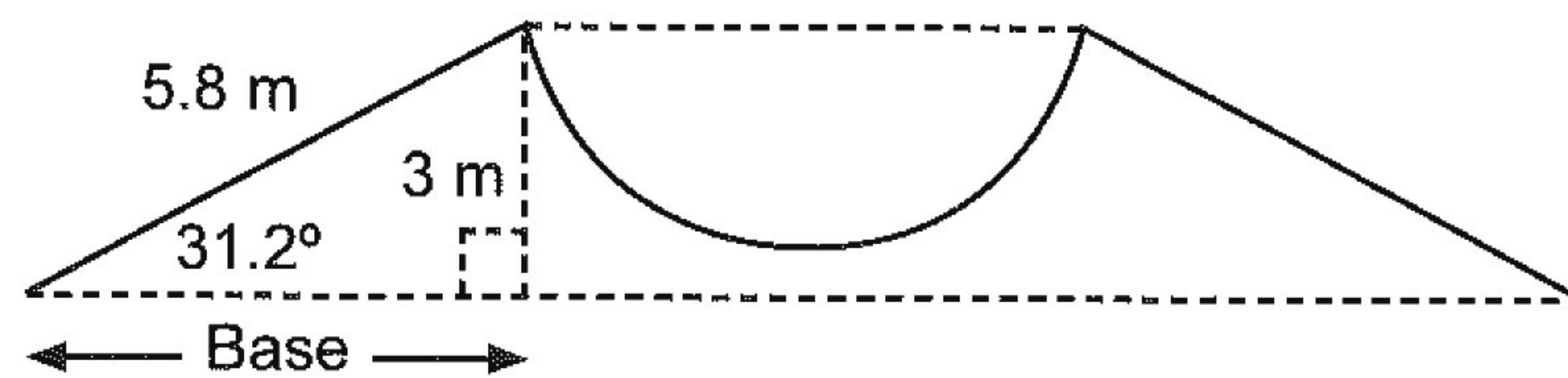
$$12.852 \div 0.162 = 79 \text{ boxes}$$

$$0.6 \times 0.9 \times 0.3 \quad 0.162 = \text{box each}$$

Question 4

(14 marks)

Below is a cross-section of a skateboard half-pipe with ramps.



The designer calculated the base of the left ramp to be 6.5 m.

$$b^2 = 3^2 + 5.8^2$$

$$b^2 = 9 + 33.64$$

$$b^2 = 42.64$$

$$b = 6.5 \text{ m}$$

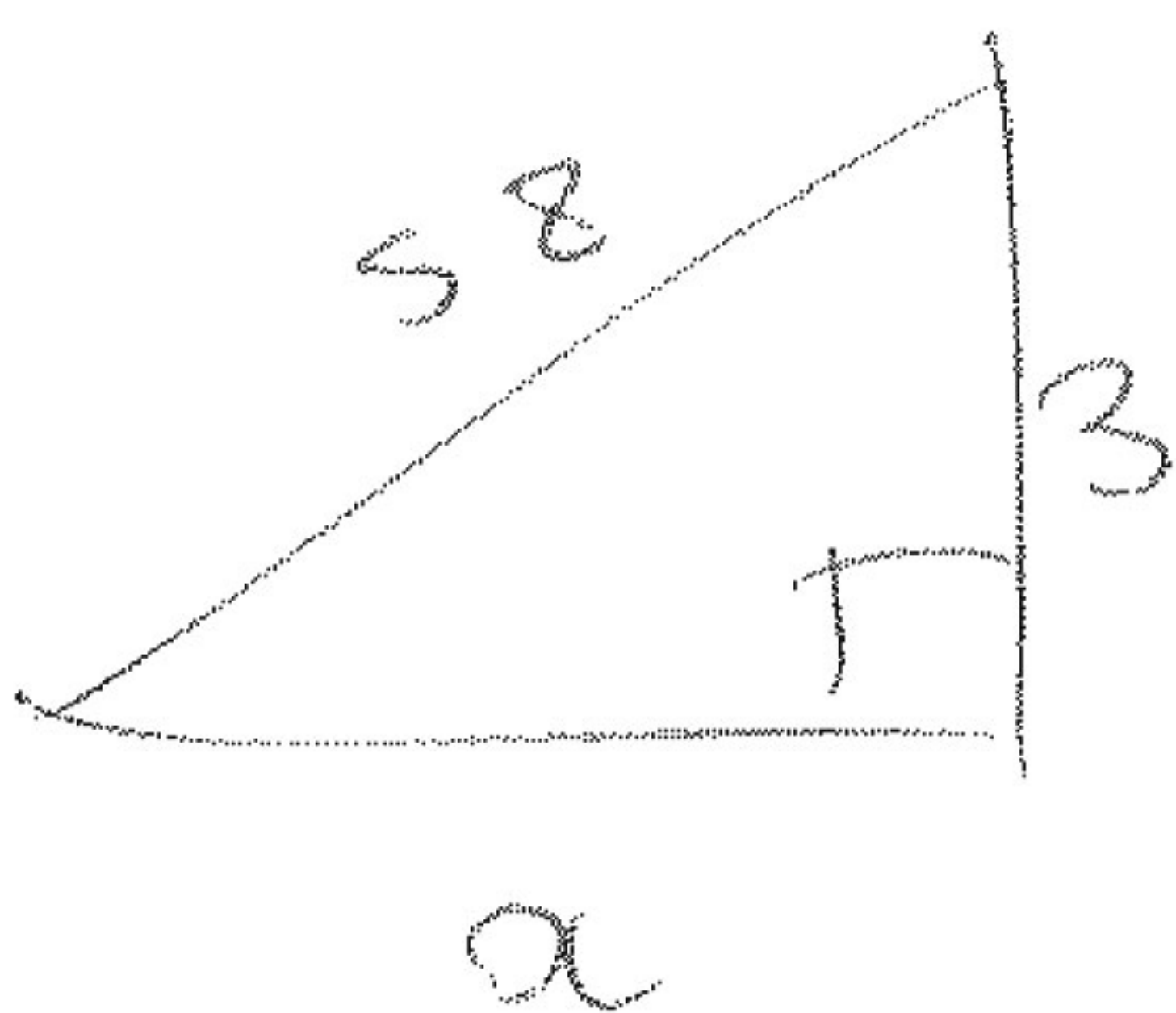
- (a) (i) Why is this answer impossible?

(1 mark)

Because it is larger than the hypotenuse

- (ii) Determine the length of the base.

(4 marks)



$$a^2 = c^2 - b^2$$

$$a^2 = 5.8^2 - 3^2$$

$$a = \sqrt{33.64 - 9}$$

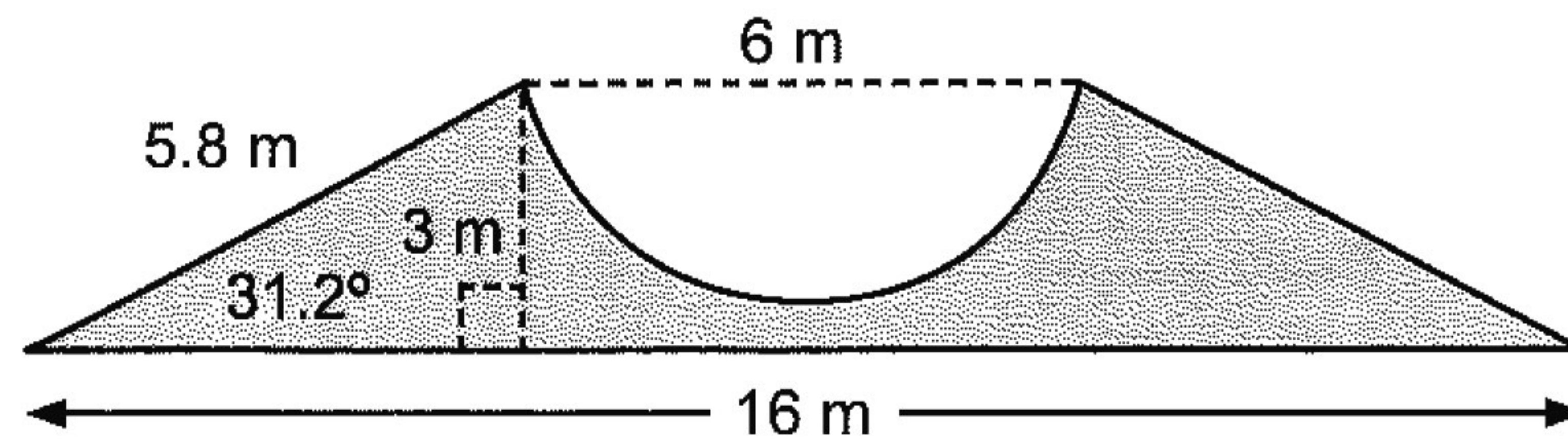
$$a = \sqrt{24.64}$$

$$a = 4.96 \text{ m}$$

- (b) The design involves a half-circle with a diameter of 6 m.

(3 marks)

Show that the area of the cross-section is 18.9 m².



$$\frac{1}{2} \times 3 \times (6 + 16) = 33$$

$$\frac{1}{2} \times \pi \times 3^2 = 14.14$$

$$33 - 14.14 = 18.86 = 18.9 \text{ m}^2$$

$$33 - 14$$

28

14

$$\frac{1}{2} \times 3 \times (6 + 16)$$

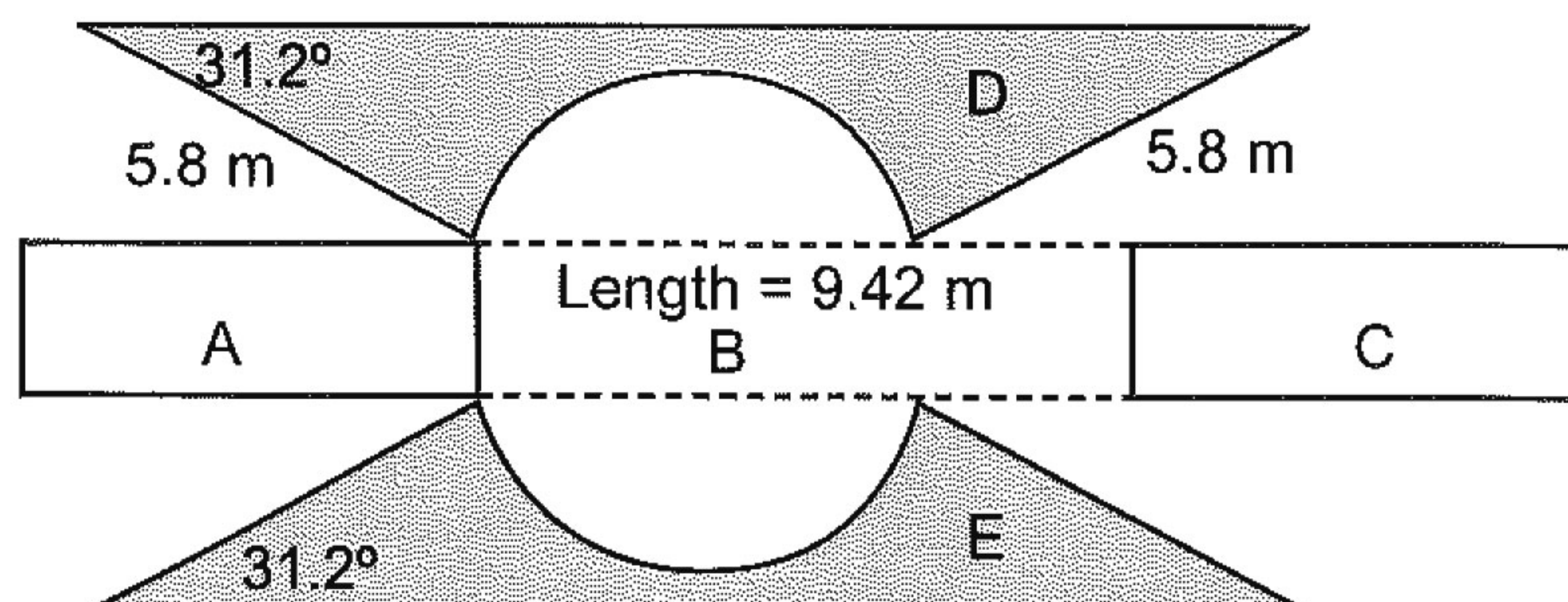
33

$\pi \times 3^2$

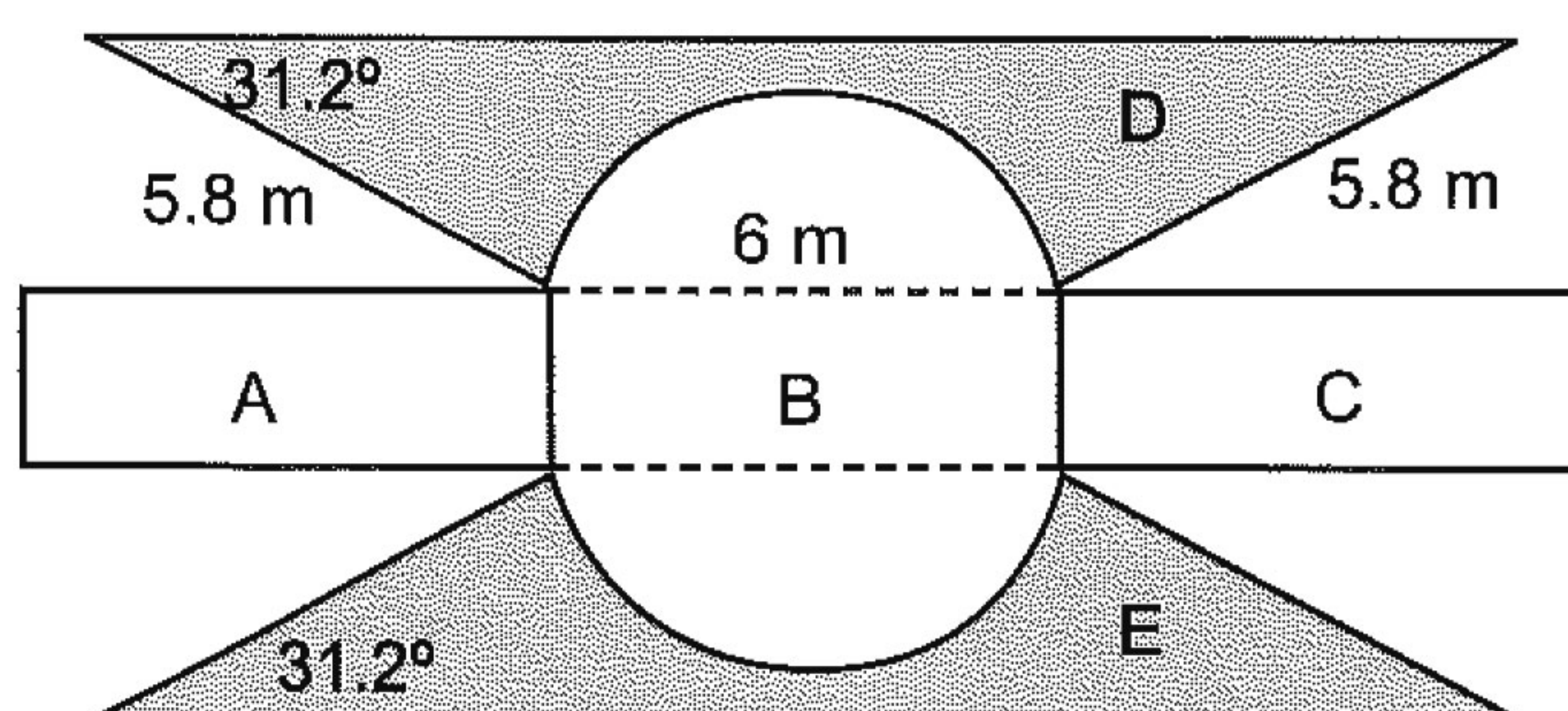
The ramp has a width of 3 m.

The designer produced two nets for the design of the ramp.

Net 1



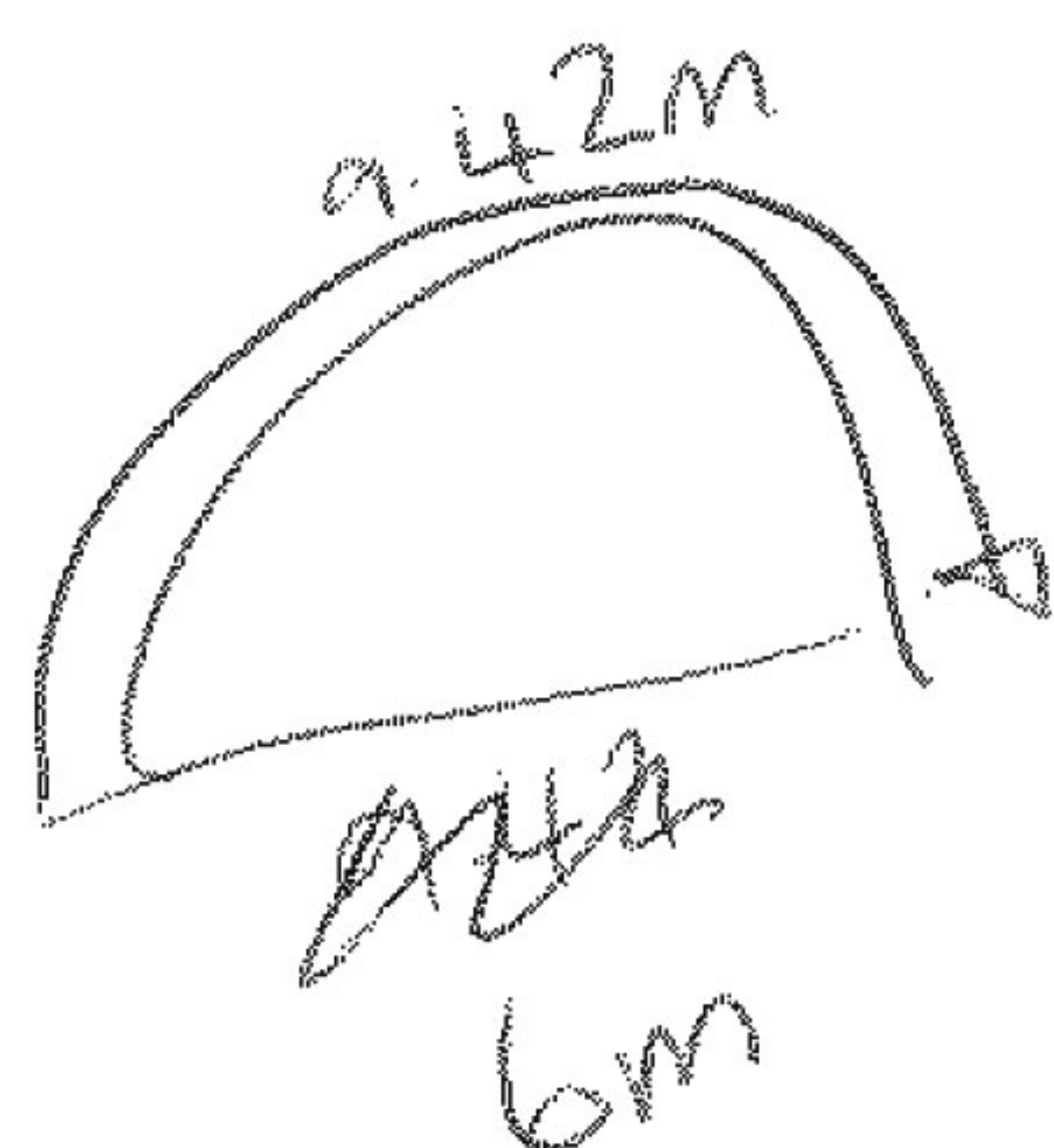
Net 2



- (c) (i) Which of these two nets would you use to construct the ramp? (1 mark)

Net 1

- (ii) Justify your choice mathematically. (1 mark)



$$6 \div 2 = r$$

$$\pi \times r = B$$

$$\pi \times 3 = 9.42$$

- (d) Using your answer in parts ^B(a) and ^C(b), find the surface area of the skateboard half-pipe and ramps. (4 marks)

$$9.42 \times 3 = 28.26 \text{ m}^2$$

$$18.9 + 18.9 = 37.8 \text{ m}^2$$

$$(5.8 \times 3) \times 2 = 34.8 \text{ m}^2$$

$$= 100.86 \text{ m}^2$$